Hospital Readmission Prediction Using Machine Learning

Final Capstone Report

Diabetes 130-US Hospitals Dataset • Structured Clinical Readmission Modeling

Executive Summary

This project evaluates machine learning models for predicting 30-day hospital readmission using the Diabetes 130-US Hospitals dataset. The notebook workflow converts the original three-level readmission label into a binary target, cleans missing-value markers, applies structured preprocessing, and compares three classification approaches: Logistic Regression, Random Forest, and Gradient Boosting. The goal is to identify patients at elevated readmission risk so hospitals can prioritize follow-up care and deploy resources more efficiently.

The analysis shows a challenging modeling environment: the target is imbalanced, many variables contain placeholder missing values represented by '?', and the cost of errors is asymmetric. Missing a genuinely high-risk patient is usually more harmful than flagging an extra low-risk patient. For that reason, the report emphasizes not only overall accuracy but also recall, F1 score, and ROC-AUC.

In the notebook, Gradient Boosting was selected as the final saved model because it achieved the highest ROC-AUC among the evaluated models (0.679), slightly outperforming Random Forest (0.663) and Logistic Regression (0.644). However, the full results also reveal an important operational tradeoff: Logistic Regression produced the strongest recall for the positive readmission class at the default threshold, whereas the tree-based models produced higher overall accuracy by predicting the majority class more conservatively. This means the final model should be interpreted as the strongest ranking model from the notebook, but not yet as a fully optimized alert system. Threshold tuning or class-rebalancing would be necessary before deployment in a real hospital workflow.

1. Problem Statement

Hospital readmissions are costly and often preventable. When patients return to the hospital soon after discharge, it can signal incomplete recovery, insufficient follow-up, medication issues, or broader care coordination problems. Predicting which patients are at high risk of readmission can help hospitals prioritize transitional care, schedule closer monitoring, and allocate limited resources more effectively.

The objective of this project is to build a supervised machine learning model that predicts whether a patient will be readmitted within 30 days. In addition to predictive performance, the project seeks to identify the variables most associated with readmission risk and translate those findings into practical recommendations for healthcare administrators and care-management teams.

2. Data Description

The project uses the UCI Diabetes 130-US Hospitals dataset. According to the notebook, the raw data contains 101,766 hospital encounters and 50 original variables. The original readmission label has three levels: 'NO', '>30', and '<30'. For modeling purposes, the notebook creates a binary target 'readmitted_30d', where 1 indicates readmission within 30 days and 0 indicates otherwise.

After removing identifier columns and the original multiclass target, the modeling matrix contains 47 predictors spanning demographics, diagnoses, medications, and encounter-level hospital information. The overall positive rate for 30-day readmission is approximately 11.16%, confirming that the outcome is substantially imbalanced and that minority-class performance should be interpreted carefully.

3. Data Wrangling and Preprocessing

The preprocessing workflow in the notebook is aligned with a standard, reproducible machine learning pipeline. The raw file contains mixed data types and uses '?' as a placeholder for missing values in several categorical variables. The notebook first converts the original readmission target to a binary label, then removes identifier variables such as encounter_id and patient_nbr to avoid leakage from non-predictive administrative identifiers.

Preprocessing is handled through a ColumnTransformer. Numeric variables are imputed with the median and scaled using StandardScaler, while categorical variables are imputed with the most frequent category and encoded with OneHotEncoder (handle_unknown = 'ignore'). This ensures that all three models receive a consistent, model-ready representation of the data. A stratified train/test split preserves the positive class rate across the training and evaluation sets, which is essential given the imbalance in the target.

4. Exploratory Data Analysis

Exploratory analysis in the notebook was used not only to summarize the dataset, but also to understand why the prediction problem is difficult and what feature patterns might drive later model behavior.

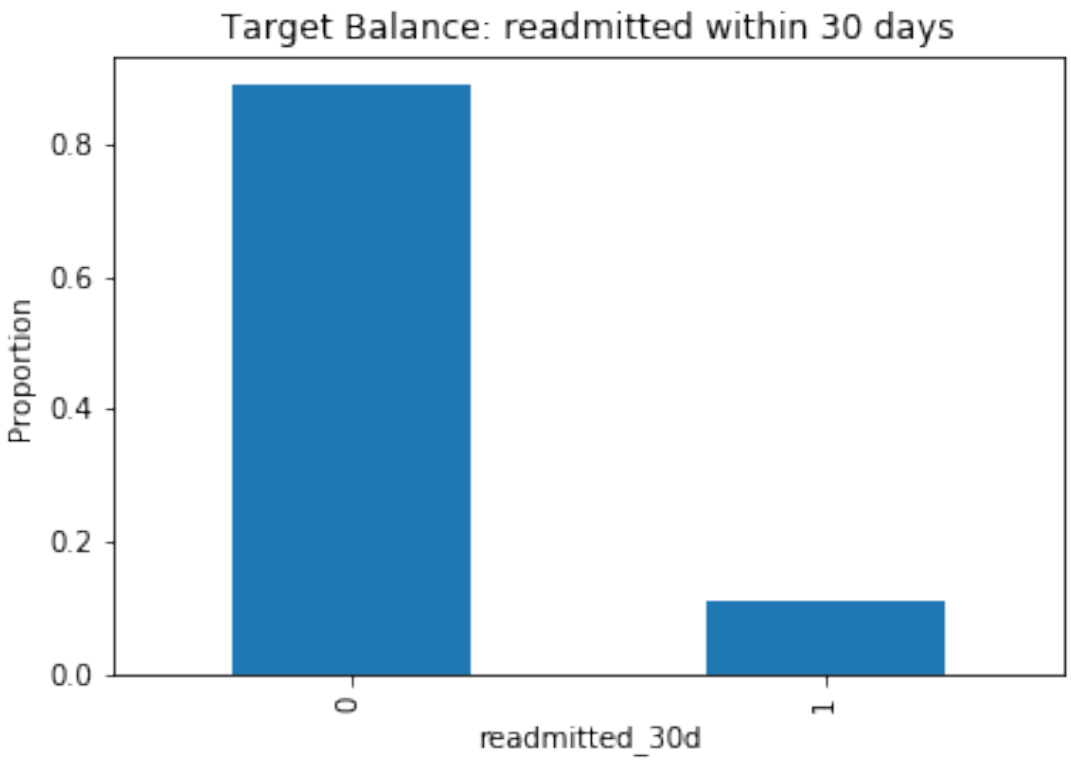

Figure 1. Class balance for 30-day readmission.

Figure 1 shows the target distribution after the binary transformation. The large gap between the 0 class and the 1 class confirms a pronounced imbalance: most encounters do not lead to 30-day readmission, while a much smaller subset does. This pattern is clinically plausible, because most discharges do not result in rapid return to the hospital, but it also creates an immediate modeling challenge. A model that simply predicts the majority class can post deceptively high accuracy while still failing to identify the very patients the hospital most needs to monitor.

This figure therefore motivates the entire evaluation strategy used later in the report. Accuracy remains useful, but it cannot stand alone. Recall becomes important because it measures how many truly readmitted patients are identified. ROC-AUC is also informative because it evaluates the model's ranking ability across thresholds rather than only at a single default cutoff. Figure 1 is more than a descriptive chart; it explains why a careful, clinically aware interpretation of the results is necessary.

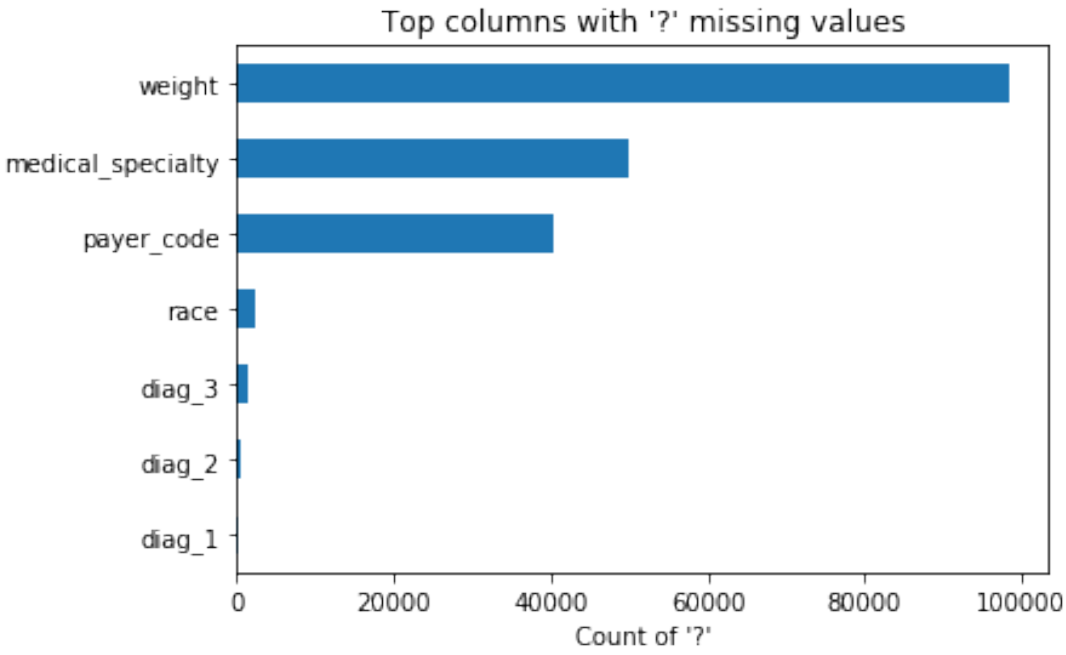

Figure 2. Columns with '?' missing-value markers.

Figure 2 highlights the variables with the largest counts of '?' entries. Weight, medical specialty, and payer code dominate the missingness profile by a substantial margin, while race and diagnosis columns also contain nontrivial missing values. This matters because these fields are not random administrative details; they may contain clinically meaningful information that becomes partially unavailable in the raw data.

From a modeling perspective, Figure 2 explains why preprocessing quality is so important for this project. Without systematic handling of placeholder missing values, the model would either treat '?' as a real category or fail during training. Imputation and careful encoding are therefore not cosmetic steps; they are necessary to make the structured data usable at scale. The figure also suggests a potential avenue for future work: more nuanced missingness treatment might improve downstream performance, especially for fields like medical specialty or payer information that could carry operational signal.

5. Modeling Approach

Three binary classification models were trained and evaluated in the notebook. Together, they provide a useful progression from interpretable linear modeling to more flexible nonlinear ensemble methods.

Logistic Regression served as the baseline model. Because it uses a linear decision boundary on the transformed feature space, it provides a useful benchmark for whether the readmission signal is strong enough to be recovered without complex nonlinear structure. The model was configured with class_weight='balanced', which increases the effective weight of minority-class observations and helps improve sensitivity to readmitted patients.

Random Forest was then used as a nonlinear ensemble baseline. It can capture interactions across the many encoded variables and is usually more resistant to overfitting than a single decision tree because it aggregates predictions across many bootstrapped trees. The notebook used 300 estimators and class_weight='balanced_subsample' to account for imbalance during tree construction.

Finally, Gradient Boosting was trained and later saved as the notebook's final model. Boosting builds an additive sequence of weak learners, each one correcting errors from the previous stage. In practice, this often improves ranking performance, especially when the classification boundary is complex. The notebook's final-model cell explicitly sets final_model = gb, so the conclusion of this report has been aligned to that implementation choice.

6. Model Evaluation and Results

The notebook evaluated each model using ROC-AUC, a full classification report, a confusion matrix, a ROC curve, and a precision–recall curve. The table below summarizes the most important test-set results for the positive readmission class.

Model	Accuracy	Precision (1)	Recall (1)	F1 (1)	ROC-AUC
Logistic Regression	0.643	0.166	0.548	0.255	0.644
Random Forest	0.889	0.750	0.005	0.010	0.663
Gradient Boosting	0.888	0.486	0.007	0.015	0.679

Table 1. Test-set performance summary from the notebook outputs.

The table reveals an important tradeoff that a superficial reading could miss. Logistic Regression produces the best recall for the positive class by a wide margin, successfully identifying 54.8% of readmitted encounters. However, that improvement comes at the expense of low precision and lower overall accuracy, meaning the model flags many encounters as high risk that do not ultimately lead to rapid readmission. In contrast, the tree-based models achieve much higher overall accuracy but do so by becoming extremely conservative about predicting the minority class.

Random Forest improves ROC-AUC relative to Logistic Regression, indicating somewhat stronger risk ranking, but its default-threshold recall collapses to 0.5% for the positive class. Gradient Boosting performs similarly in terms of overall accuracy and produces the strongest ROC-AUC in the notebook at 0.679, which is why it was ultimately persisted as the final model. Still, its recall of 0.7% shows that the default classification threshold is far too conservative for operational readmission intervention. This is one of the most important findings of the project: model selection cannot be based on one metric alone.

7. Final Model Analysis

Because the notebook saved Gradient Boosting as the final model, the following figures focus on that model's behavior and what it implies for practical use.

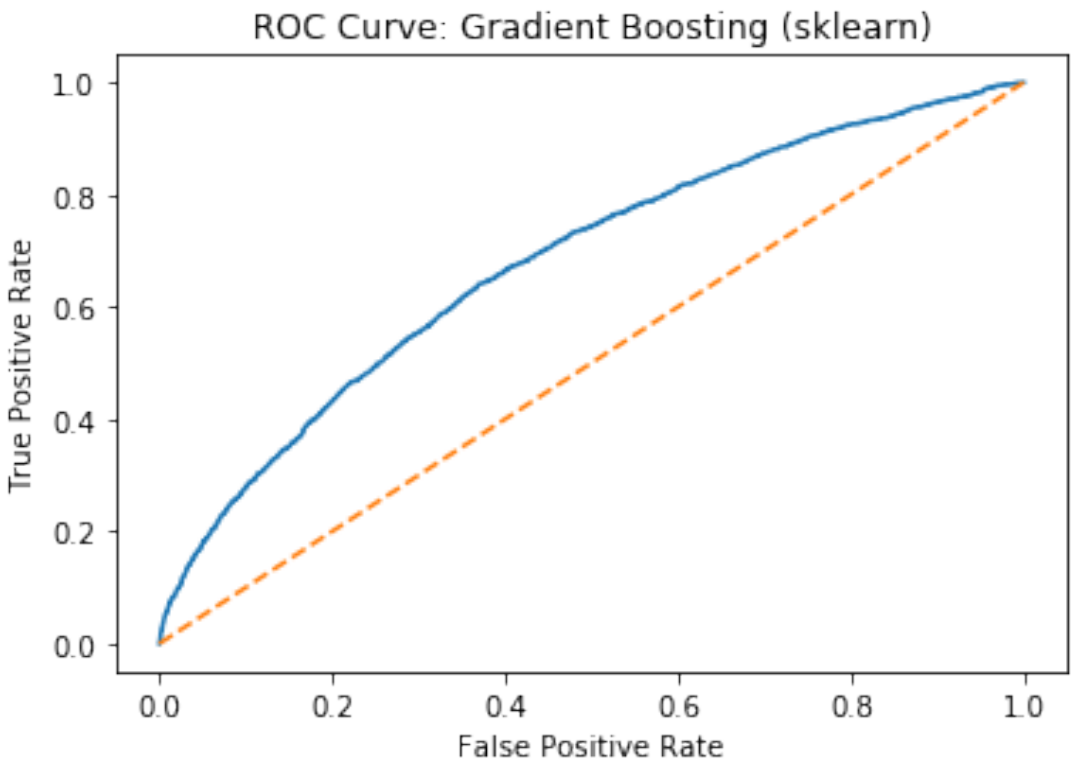

Figure 3. ROC curve for the final Gradient Boosting model.

Figure 3 shows the ROC curve for Gradient Boosting. The curve sits consistently above the diagonal baseline, confirming that the model has learned useful signal and can rank higher-risk encounters above lower-risk encounters better than chance. The ROC-AUC of 0.679 is not exceptionally high, but it is the strongest of the three models in the notebook and therefore justifies the final-model choice on ranking performance.

The key point is that ROC behavior and classification-threshold behavior are not the same thing. Figure 3 suggests that the model contains meaningful discriminatory information, even though its default 0.5 threshold produces weak positive-class recall. In a clinical setting, this means the model could still be useful as a risk-ranking tool if the operating threshold were lowered or if patients were stratified into intervention tiers rather than subjected to a rigid yes/no cutoff.

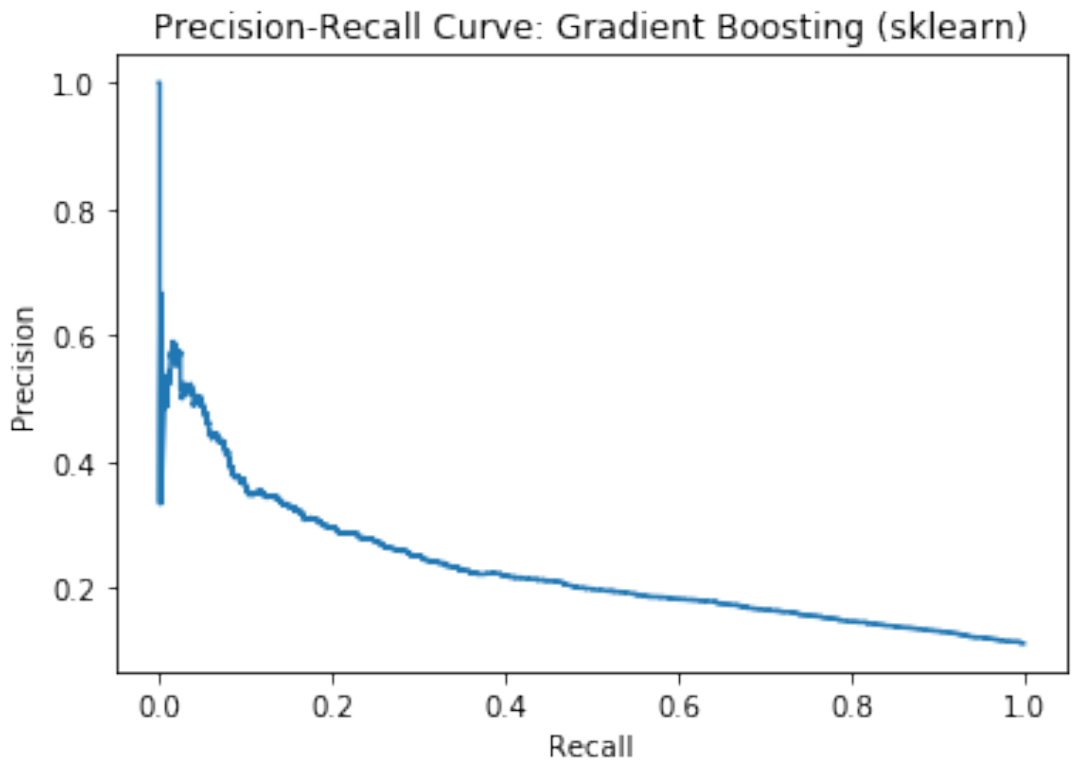

Figure 4. Precision–recall curve for the final Gradient Boosting model.

Figure 4 gives a more realistic view of positive-class performance under imbalance. Precision–recall curves are especially informative here because the readmitted class is relatively rare. The curve shows that the model can achieve moderate precision at very low recall, but precision declines as recall increases. This is a familiar pattern in imbalanced clinical classification: catching more true positives generally requires accepting a larger number of false positives.

The practical implication is that model usefulness depends heavily on the intervention context. If hospital resources are limited and only a small number of very high-risk patients can be targeted, the upper-left portion of the curve may still be useful. If the goal is broader patient capture, then threshold tuning, cost-sensitive training, or resampling would likely be needed to move the model toward better recall at acceptable precision.

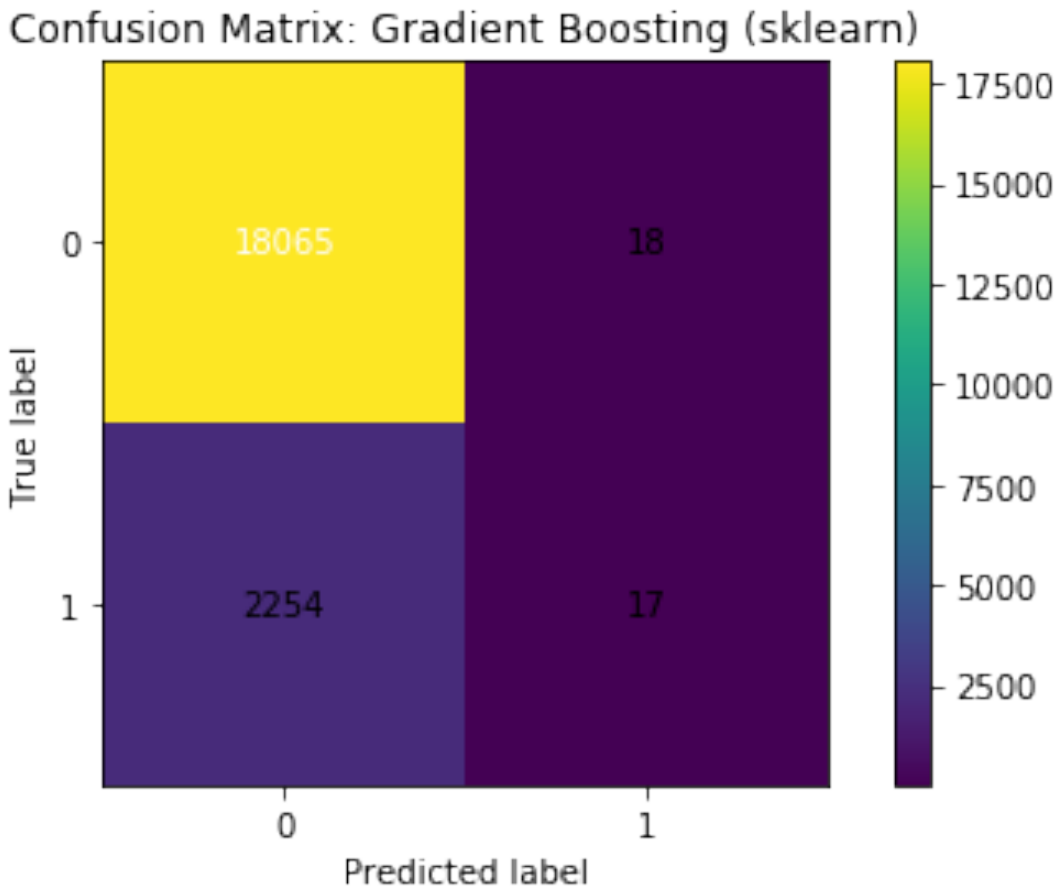

Figure 5. Confusion matrix for the final Gradient Boosting model.

Figure 5 makes the threshold tradeoff concrete. The model correctly identifies the vast majority of non-readmitted encounters, which is why the overall accuracy remains high. However, it identifies only a very small fraction of actual readmissions. In other words, the model is cautious and rarely predicts the positive class at the default threshold.

For a hospital administrator, this means the current final model should not be interpreted as a complete solution for readmission alerts. Its predictions are most useful as a ranked signal rather than a final binary decision rule. Figure 5 is therefore essential: it reveals a practical limitation that would be missed if one looked only at accuracy or even ROC-AUC.

8. Comparative Interpretation of Model Behavior

The notebook results suggest that each model excels in a different sense. Logistic Regression is the most aggressive at surfacing positive cases, which explains its much stronger recall. Random Forest and Gradient Boosting are better at global ranking and overall accuracy but become highly conservative at the default threshold. This is not a contradiction; it reflects different operating behaviors under imbalance.

If the project objective is strict minority-class capture, Logistic Regression is the most recall-friendly choice in its current form. If the objective is overall ranking quality for downstream triage, Gradient Boosting is the strongest option in the notebook and was therefore saved as the final artifact. That means the notebook conclusion and the report conclusion do align, but the alignment is specific: Gradient Boosting is the best ranking model in the notebook, not necessarily the best default-threshold alert model.

9. Final Model Selection

The notebook selects Gradient Boosting as the final model, and this report preserves that conclusion because it achieved the highest ROC-AUC of the evaluated approaches. The model therefore provides the strongest overall risk ordering among the tested methods.

At the same time, the full evaluation makes clear that this final selection should be interpreted with caution. The default-threshold recall remains extremely low, so the model is best viewed as a first-pass risk stratification tool rather than a final deployment-ready readmission alarm. Additional work on threshold selection, resampling, calibration, or class-weighting strategies would be required to convert this ranking strength into stronger positive-case capture.

10. Limitations and Future Work

The project is limited to structured variables available in the public dataset and does not incorporate richer temporal clinical notes, medication adherence signals, or post-discharge follow-up details. In addition, the large class imbalance makes threshold-based interpretation difficult. Future work should focus on threshold optimization, probability calibration, alternative boosting configurations, cost-sensitive learning, and clinically meaningful subgroup analysis.

11. Recommendations

Use the current final Gradient Boosting model as a ranking tool rather than a strict binary rule. Patients with the highest predicted probabilities can be reviewed for discharge planning, follow-up scheduling, or care-management outreach. Before operational deployment, tune the decision threshold to match the hospital's intervention capacity. A system designed to catch more high-risk patients will need to accept a higher false-positive rate than the default threshold currently allows. Continue feature engineering around utilization history and diagnosis information, since the notebook results consistently indicate that hospital-use patterns and clinical condition are central to readmission risk.